

A Boundary Prime–Power Decomposition and Local Perron Residues for Dirichlet L -Functions

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Proof and novelty audit — 1 August 2026

Abstract

We record a boundary prime-power decomposition and the exact local residue of a shifted Perron kernel. The boundary identity is unconditional and, in fact, does not use that the evaluation point is a zero: its only delicate term is an ordinarily convergent prime sum on $\Re z = 1$. We fix the logarithm by a right-hand Euler-product limit, so no false path-independence assertion is needed. The residue is a local Laurent calculation and has been formalized in Lean 4. Neither result supplies a global partial-Möbius-sum asymptotic: off-target zeros contribute non-decaying oscillatory residues on the Riemann hypothesis. A targeted literature audit finds the analytic ingredients in earlier work of Conrad and Akatsuka; no novelty claim is made for the decomposition itself.

Status. This is a proved technical note, not a submission draft. The theorem and local residue below are complete at paper level. The independent novelty gate fails for a standalone research paper because the theorem is an elementary specialization of established boundary Euler-product machinery. No global c_K asymptotic is claimed.

1 Setup and branch convention

Let χ be a Dirichlet character modulo q , let $s = \frac{1}{2} + i\tau$ with $\tau \neq 0$, and let ψ be the primitive character of conductor $f \mid q$ that induces the (possibly imprimitive) character χ^2 . Define

$$T_K(\chi, s) := \sum_{p \leq K} \sum_{k \geq 2} \frac{\chi(p)^k}{kp^{ks}}. \quad (1)$$

For $z = 1 + 2i\tau$, define the Euler-product-normalized boundary logarithm

$$\text{Log}_* L(z, \psi) := \lim_{\epsilon \downarrow 0} \sum_p \sum_{r \geq 1} \frac{\psi(p)^r}{rp^{r(z+\epsilon)}}. \quad (2)$$

The limit exists by the boundary Abel theorem used below. This is a branch *definition*, not a claim of path independence. Continuation around a zero can change a logarithm by $2\pi i$, so arbitrary-path independence would be false.

Set

$$\text{BPC}_1(\chi, s) := \frac{1}{2} \sum_{\substack{p \mid q \\ p \nmid f}} \text{Log}(1 - \psi(p)p^{-2s}), \quad (3)$$

$$\text{BPC}_2(\chi, s) := -\frac{1}{2} \sum_{r \geq 2} \frac{1}{r} \sum_p \frac{\chi(p)^{2r}}{p^{2rs}}, \quad (4)$$

$$T_{\geq 3}(\chi, s) := \sum_{k \geq 3} \frac{1}{k} \sum_p \frac{\chi(p)^k}{p^{ks}}. \quad (5)$$

The last two double sums are absolutely convergent. In BPC_1 , $|\psi(p)p^{-2s}| = p^{-1} < 1$, so $\text{Log}(1 - w) = -\sum_{r \geq 1} w^r / r$ unambiguously denotes the principal logarithm.

2 Boundary identity

Theorem 1 (Boundary prime-power decomposition). *Under the setup above, the ordinary prime-cutoff limit exists and*

$$\lim_{K \rightarrow \infty} T_K(\chi, s) = \frac{1}{2} \text{Log}_* L(2s, \psi) + \text{BPC}_1(\chi, s) + \text{BPC}_2(\chi, s) + T_{\geq 3}(\chi, s). \quad (6)$$

Proof. For every finite K , absolute convergence of the inner series gives

$$T_K(\chi, s) = \frac{1}{2} \sum_{p \leq K} \frac{\chi(p)^2}{p^{2s}} + \sum_{p \leq K} \sum_{k \geq 3} \frac{\chi(p)^k}{kp^{ks}}. \quad (7)$$

The second term converges absolutely as $K \rightarrow \infty$, since

$$\sum_p \sum_{k \geq 3} \frac{|\chi(p)|^k}{kp^{k/2}} \leq \frac{1}{3(1 - 2^{-1/2})} \sum_p p^{-3/2} < \infty.$$

Thus no conditional rearrangement is being made.

Put $\vartheta = \chi^2$, viewed as a character modulo q . Conrad's boundary Dirichlet-series theorem [2, Theorem 4.1], together with his comparison of prime and prime-power cutoff orders [2, Lemma 3.1], applied to the Euler-product logarithm of $L(z, \vartheta)$ shows that, for $z = 1 + 2i\tau$ with $\tau \neq 0$, the prime-cutoff series

$$\sum_p \sum_{r \geq 1} \frac{\vartheta(p)^r}{rp^{rz}} \quad (8)$$

converges and equals its right-hand boundary value from $\Re z > 1$. The required hypotheses follow from the classical nonvanishing of Dirichlet L -functions on $\Re z = 1$; when ϑ is principal, the only pole is at $z = 1$, excluded by $\tau \neq 0$. Akatsuka independently gives the direct principal-character estimate

$$\sum_{X < p \leq Y} p^{-1-2i\tau} = O_\tau(1/\log X) \quad (9)$$

in equations (2.3)–(2.5) of [1].

The $r \geq 2$ part of (8) is absolutely convergent. Subtracting it, without changing order, proves

$$\sum_p \frac{\chi(p)^2}{p^{2s}} = \text{Log}_* L(2s, \vartheta) - \sum_{r \geq 2} \frac{1}{r} \sum_p \frac{\chi(p)^{2r}}{p^{2rs}}. \quad (10)$$

Finally, for $\Re z > 1$ the inducing-character identity is

$$L(z, \vartheta) = L(z, \psi) \prod_{\substack{p|q \\ p \nmid f}} (1 - \psi(p)p^{-z}).$$

Take Euler-product logarithms there and then the horizontal limit to $2s$. The finite factors stay inside the disk $|w - 1| < 1$, so their principal logs agree with their convergent power series. Hence

$$\text{Log}_* L(2s, \vartheta) = \text{Log}_* L(2s, \psi) + \sum_{\substack{p|q \\ p \nmid f}} \text{Log}(1 - \psi(p)p^{-2s}).$$

Multiplying (10) by $1/2$ and substituting it in (7) proves (6). $\square$

Remark 2 (Weakness and convergence mode). *Theorem 1 does not require s to be a zero of $L(s, \chi)$, nor does it require simplicity. The limit is ordinary prime-cutoff convergence. Only the $k = 2$ term is conditional; all other passages use absolute convergence. The radial limit in (2) agrees with the ordinary boundary series by Abel's theorem, but arbitrary complex paths are not part of the statement.*

3 The local double-pole residue

Lemma 3 (Local Perron residue). *Let L be holomorphic near ρ , with $L(\rho) = 0$ and $L'(\rho) \neq 0$. For $K > 1$,*

$$\text{Res}_{w=0} \frac{K^w}{wL(\rho+w)} = \frac{\log K}{L'(\rho)} - \frac{L''(\rho)}{2L'(\rho)^2}. \quad (11)$$

Proof. Taylor expansion at the simple zero gives

$$\frac{1}{L(\rho+w)} = \frac{1}{L'(\rho)w} - \frac{L''(\rho)}{2L'(\rho)^2} + O(w),$$

while

$$\frac{K^w}{w} = \frac{1}{w} + \log K + O(w).$$

The coefficient of w^{-1} in their product is the claimed residue. $\square$

Remark 4 (Formal scope). *Lemma 3 is represented by a zero-sorry Lean 4 theorem for an analytic function with a simple zero at the origin. The formal result certifies the local Laurent algebra; it does not certify a Perron contour shift or Theorem 1.*

4 Why no global partial-sum asymptotic is claimed

Applying a truncated Perron formula to

$$c_K(\chi, \rho) = \sum_{n \leq K} \frac{\mu(n)\chi(n)}{n^\rho}$$

and shifting the contour crosses every off-target zero ρ' . In the simple zero case its contribution contains a factor of the shape

$$\frac{K^{\rho'-\rho}}{(\rho' - \rho)L'(\rho', \chi)}. \quad (12)$$

Under RH, $\Re(\rho' - \rho) = 0$, so this is oscillatory with modulus independent of K ; it is not automatically $o(1)$. Inoue's explicit formula [3] organizes these zero contributions but does not permit them to be discarded. A complete contour displacement would also have to specify the zero-sum ordering, bound the horizontal and left-edge integrals, and handle multiple zeros. None of those global steps is used in Theorem 1. Consequently Lemma 3 identifies one local term, not the asymptotic of c_K .

5 Numerical audit design

A reproducibility computation can use four fixed character/zero pairs and report, for every cutoff K : the left side of (6); each of the four right-side components; the complex residual and working precision; direct checks of $L(\rho) = 0$, $L'(\rho) \neq 0$, and the inducing character ψ ; and executable source, raw values, software versions, and hashes. Observed rates must be separated from unconditional estimates and from RH-conditional bounds such as those in Soundararajan [5].

6 Novelty and disposition

The exact four-term display in Theorem 1 was not found verbatim in the targeted search. That does not make it a new theorem. Conrad’s general boundary framework already identifies the squared-character second moment and controls the ordering issue [2, Example 4.6 and Theorem 4.9]; Akatsuka proves the principal boundary prime-sum estimate directly; and Kaneko treats broader Dirichlet partial Euler-product asymptotics [4]. The displayed formula is a transparent isolation of the $k = 2$ term plus the standard imprimitive Euler factors. The local residue is likewise standard Laurent algebra. Therefore these two results do not presently clear a standalone-paper novelty threshold. Their defensible use is as a technical appendix and reproducibility certificate supporting a paper with a genuinely new prime-bias theorem or numerical discovery.

The remaining open work is numerical rather than theorem-closing: rerun the four character/zero examples from raw inputs, report interval/error control, and reconcile the branch convention across code and exposition. Those checks can strengthen a computational section, but they do not change the novelty verdict above.

References

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